

MultiplierBoard: Minimal Transformer for 6-bit Binary Multiplication

Participant #179 · KRAFTON AI R&D Hackathon · Round 1, Day 1

Submission: P_1 = -1, P_2 = 17,280, Acc_2 = 1.0

1. Architecture

Decoder-only transformer with sinusoidal positional encoding (free, not counted).

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Input (12 tokens) → Embedding(2, 32) + SinPE
→ [TransformerBlock × 2]
  LN → MultiHeadAttn(2 heads) → Residual
  LN → FFN(32→64→32, GELU) → Residual
→ LayerNorm → Linear(32, 2)
→ Output (12 tokens, autoregressive greedy decoding)
```

Component	Choice	Rationale
d_model	32	Smallest stable dimension (d=24 seed-unstable)
n_heads	2	Sufficient for input bit routing
n_layers	2	Minimum for multiplication (1-layer insufficient)
d_ff	64	Standard 2× ratio
PE	Sinusoidal (free)	Fixed, not counted per rules
Activation	GELU	Standard

P_2 = 17,280 parameters. Acc_2 = 1.0 (100%).

2. Problem 1-2: Training & Results

Dimension Sweep

d_model	Params	Accuracy	Seed-stable?
48	38,208	100%	Yes
32	17,280	99.89–100%	Yes (4 seeds verified)
24	9,888	74–99.78%	No — varies by seed
16	4,544	37.8%	N/A — fails

Key finding: d=32 is the stability threshold for 6-bit multiplication. Below d=32, accuracy becomes seed-dependent. d=24 ranged 74–99.78% across seeds. d=16 fails entirely.

Weight Tying Experiments

Tying	d	Params	Acc	Stable?
None (standard)	32	17,280	100%	Yes
Embed + V=Q	32	14,912	78–99.95%	No
K=Q + V=Q	32	—	~50%	No
Embed + V=Q	24	8,496	99.46%	No

VQ tying reduces params but introduces seed instability. K=Q forces symmetric attention — the model can no longer distinguish A and B operands.

Low-Rank Factorization

Rank-2 factorization of all projections: d=32 → 2,432 params, 9.3% accuracy. Insufficient capacity for multiplication's quadratic bit interactions.

Training Curve (d=32 baseline)

Loss decreases steadily. Accuracy reaches 99%+ at epoch ~90, 100% by epoch ~150. Stable convergence with no signs of overfitting.

3. Problem 1-1: Hand-Coded Approach

We decomposed multiplication as iterated shift-and-add:

$$P = A[0]*(B<<0) + A[1]*(B<<1) + \dots + A[5]*(B<<5)$$

Building Blocks

Operation	Implementation	Params
AND(a,b)	ReLU(a + b - 1)	3
XOR(a,b)	a + b - 2·AND(a,b)	~5
Carry	ReLU(sum - 1)	~3

Implementation Attempt & Fundamental Dilemma

We built a 1-layer model (d=24, 12 heads) with PE-based attention routing (10/12 heads achieved one-hot patterns) and PE annihilation in V projections for clean bit extraction.

Design	Bit Extraction	Position Info	Verdict
With residual	Contaminated by PE	Available	MLP can't compute AND
Without residual	Clean	Lost	MLP identical for all positions

The MLP's shared weights require position info to distinguish output positions, but the residual connection contaminates bit values with PE signals. Resolution requires ≥2 layers (layer 1: gather bits, layer 2: position routing + carry). Full solution estimated at ~6 hours. **P_1 = -1.**

4. Ablations & Failed Attempts

What mattered most

- 1. **d_model ≥ 32** — most critical. Below this, learning is unstable across seeds.
- 2. **n_layers = 2** — minimum for multiplication. 1-layer insufficient (empirically confirmed).
- 3. **Seed stability** — tying reduces params but introduces training variance.

What didn't work

- **Low-rank (rank=2):** 9.3% — multiplication needs full-rank projections.
- **K=Q tying:** ~50% — symmetric attention can't distinguish operands.
- **Aggressive VQ tying at d≤24:** seed-unstable (74–99%).
- **d=16 and below:** insufficient capacity (37.8%).

Key Insight

Multiplication is fundamentally harder than addition for transformers. While AdderBoard achieved 6 parameters for 10-digit addition, multiplication's quadratic bit interactions ($A_i \times B_j$ for all $i+j=k$) and multi-bit carry propagation demand significantly more capacity. The stability boundary (d=32) reflects this complexity gap.